

The system satisfies the superposition principle if whenever an input is expressed as a linear combination of such signals, the system's response is the same linear combination of the responses to the individual signals.

Example 4: Given a scalar r , define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(x) = rx$. T is called a contraction when $0 < |r| < 1$ and a dilation when $|r| > 1$. Let $r=3$, and show that T is a linear transformation.

Solution: To show that T is a linear transformation, we need to verify the following conditions given in the definition: ① $T(x+y) = T(x) + T(y)$, for all x and y in $\mathbb{R}^2$. ② $T(cx) = cT(x)$, for any scalar c and for all x in $\mathbb{R}^2$.

$$\begin{aligned} \text{① } T(x+y) &= 3(x+y) = 3x + 3y = T(x) + T(y) \\ \text{② } T(cx) &= 3(cx) = 3cx = c \cdot (3x) = c \cdot T(x) \end{aligned} \quad \left. \vphantom{\begin{aligned} \text{① } T(x+y) &= 3(x+y) = 3x + 3y = T(x) + T(y) \\ \text{② } T(cx) &= 3(cx) = 3cx = c \cdot (3x) = c \cdot T(x) \end{aligned}} \right\} \text{Follows the rules of vector arithmetic}$$

$\therefore T$ is a linear transformation.

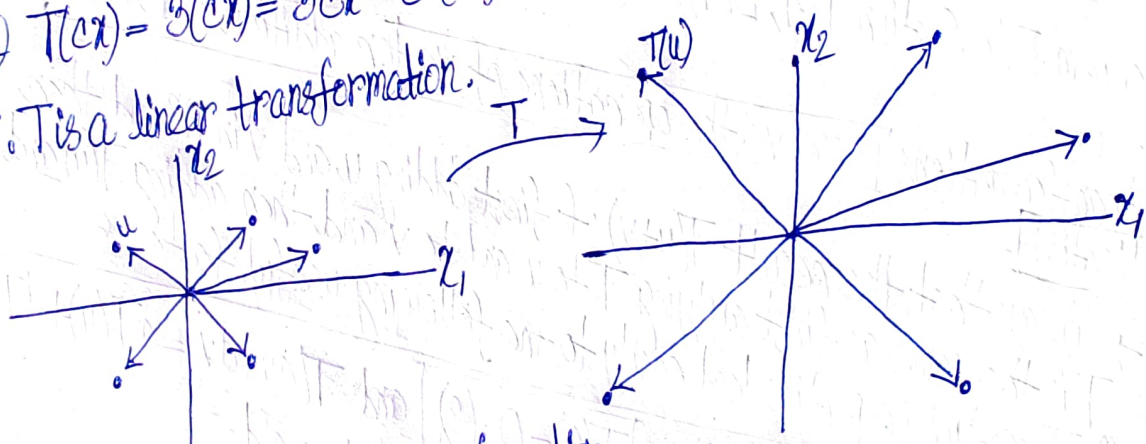


Figure 5 A dilation transformation

Example 5: Define a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(x) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$

Find the images under T of $u = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, and $u+v = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$

Solution: $T(u) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$, $T(v) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$

$$T(u+v) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{bmatrix} -4 \\ 6 \end{bmatrix}$$

Note that $T(u+v)$ is obviously equal to $T(u) + T(v)$. It appears from Figure 6 that T rotates u, v and $u+v$ counter clockwise about the origin through 90° . In fact, T transforms the entire parallelogram determined by u and v into the one determined by $T(u)$ and $T(v)$. (See Exercise 28)